A/B Test Hypothesis Document – GaleCart

Problem Statement

A significant drop-off occurs at the final step of the mobile checkout process, where first-time visitors abandon their purchase before placing an order. This reduces completed purchases, lowers revenue, and limits GaleCart's ability to improve merchant conversion performance. The experiment focuses on optimizing the checkout call-to-action (CTA) and trust-message treatment to reduce abandonment.

Experiment Scope

Audience: First-time mobile visitors aged 25–44 who added at least one item to cart but have not completed checkout in the last 30 days.
Funnel stage: Final checkout step before order placement.
Placement: Checkout CTA button and trust-message area beneath the CTA.

Audience and Funnel Definition

Only eligible first-time mobile shoppers entering the final checkout page are included. The test applies exclusively to the checkout completion stage.

Primary Hypothesis

If GaleCart replaces the existing checkout CTA with a clearer, action-oriented CTA and strengthens the trust message with a security reassurance directly beneath the CTA, then first-time mobile visitors will be more likely to complete checkout, resulting in a measurable increase in mobile checkout completion rate compared with the current experience.

Rationale

Users often hesitate at the final payment step due to uncertainty or lack of confidence. A more explicit CTA combined with a visible trust message can reduce perceived risk and encourage completion.

Variants

Control: Current 'Place Order' CTA with standard trust message.
Variant: More prominent action-oriented CTA (e.g., 'Complete Secure Order') plus enhanced trust message emphasizing secure checkout and data protection.

Metrics

Primary: Mobile checkout completion rate (orders completed / eligible checkout sessions).
Secondary: CTA click-through rate; checkout abandonment rate; average checkout completion time.
Guardrail: Payment error rate; page load time; refund/cancellation rate within 24 hours.

Assumptions and Exclusions

Assumptions: Stable mobile traffic, consistent tracking, equal random assignment, no major promotional changes during the test.
Exclusions: Desktop/tablet users, returning purchasers, internal traffic, bots, users outside the target age range when identifiable.

Decision Criteria

Success: Statistically significant improvement in checkout completion without negative guardrail impact.
Inconclusive: No statistically significant difference or insufficient sample size.
Reject: Variant performs worse than control or negatively affects guardrail metrics.